RSA Public-Key Cryptosystem

24. Write a C program for RSA system, the public key of a given user is e = 31, n = 3599. What is the private key of this user? Hint: First use trial-and-error to determine p and q; then use

Aim

To write a C program to determine the private key corresponding to the RSA public key \(e=31,\ n=3599\) using prime factorization and the Extended Euclidean Algorithm.

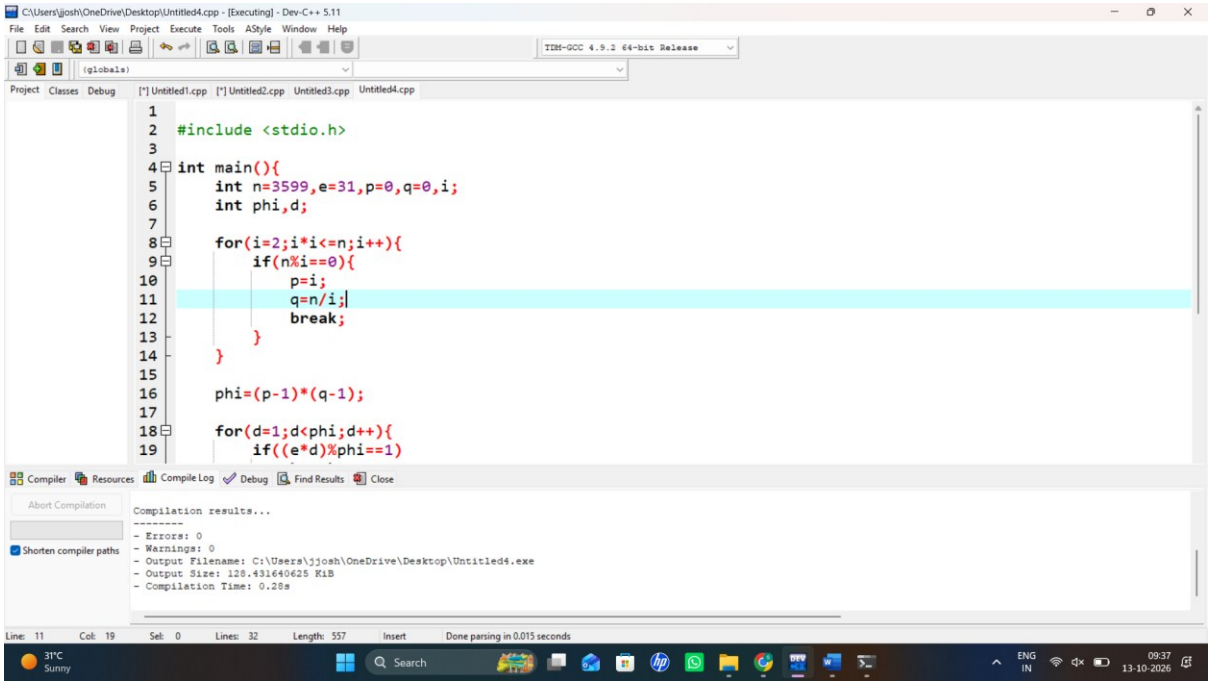
Algorithm

1. Start the program.
2. Read the public key values \(e=31\) and \(n=3599\).
3. Find the prime factors \(p\) and \(q\) of \(n\).
4. Calculate Euler's totient function:

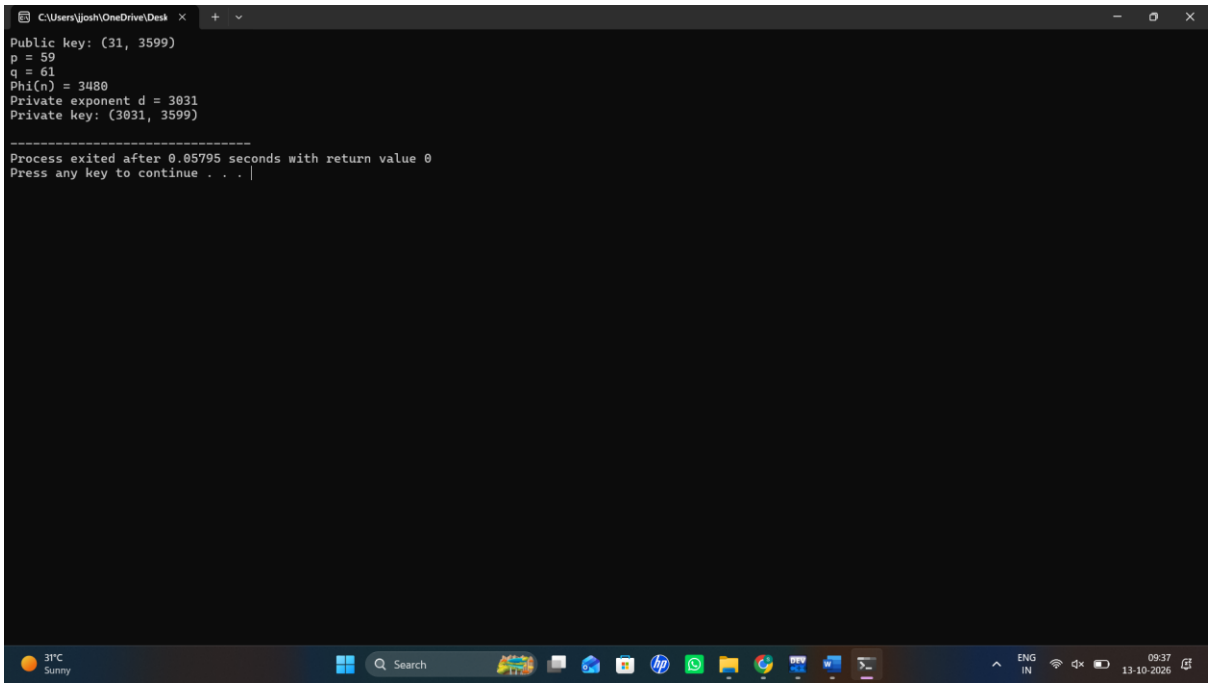
$$\phi(n)=(p-1)(q-1)$$

5. Find the multiplicative inverse \(d\) of \(e\) modulo \(\phi(n)\).
6. Display the private key \((d,n)\).
7. Stop the program.

Program :



Output:



Result

Thus, the RSA private key was calculated successfully.

The prime factors are \(p=59\) and \(q=61\).

\[\phi(n)=(59-1)(61-1)=3480 \]

The private exponent satisfies

\[31 \times 3031 \equiv 1 \pmod{3480} \]

Therefore, the private key is \((3031,3599)\).